

Working Group VII - Dark Matter

Focus: The effects of baryonic physics on the properties of the simulated galaxies' dark matter halos

Project Leaders: **Miguel Rocha** & Michael Kuhlen

Participants: Mike Boylan-Kolchin, Javiera Guedes, Sam Leitner, Piero Madau, Annalisa Pillepich, Joel Primack, Justin Read, Romain Teyssier, Sebastian Trujillo-Gomez

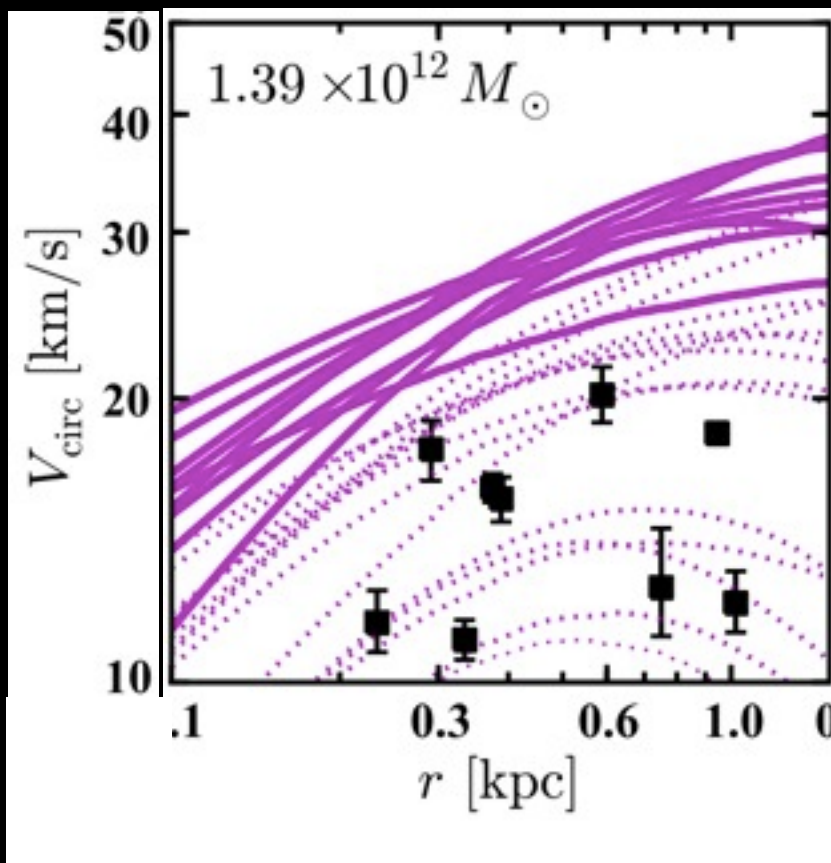
AGORA Workshop
08/17/2013

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Milky Way dwarfs

($\sim 10^9$ – $10^{10} M_{\odot}$ halos)



“Too big to fail”

(Boylan–Kolchin+ 2011)

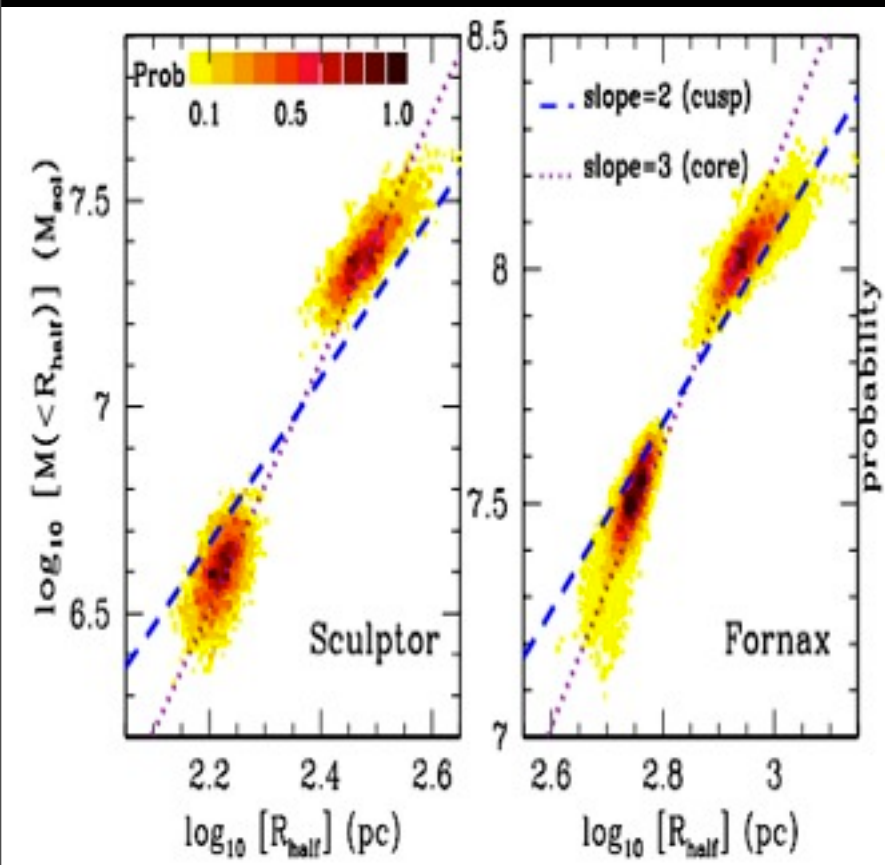
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Observed Cores in ~ 0.5 kpc

(Walker & Penarrubia 2011,
but see Strigari et al. 2010)

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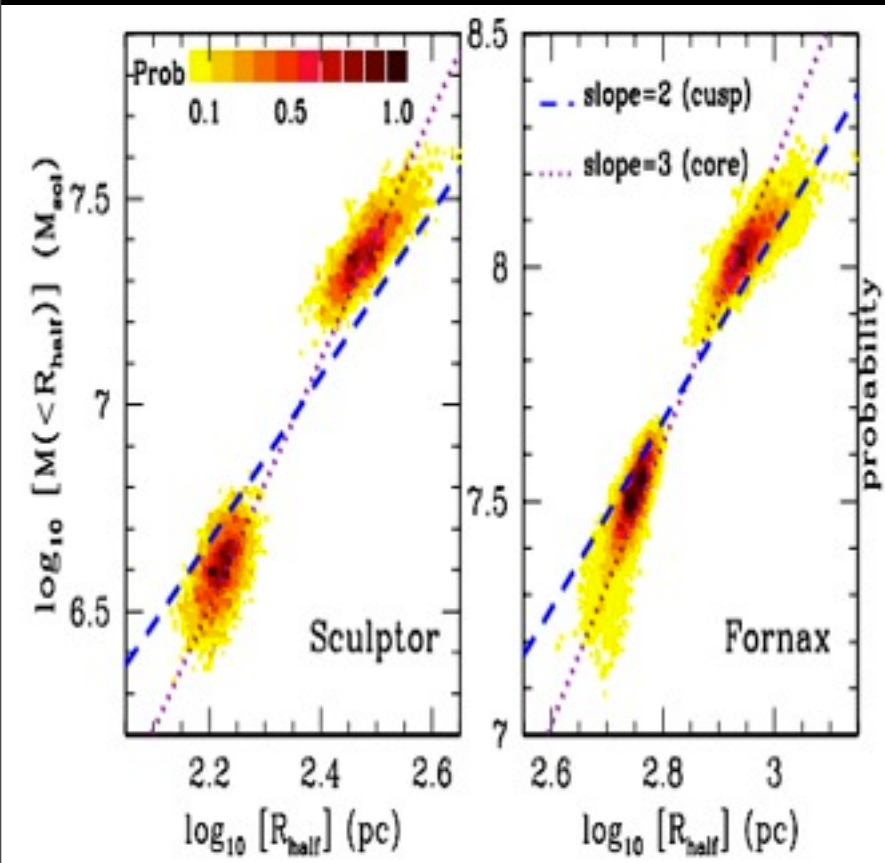
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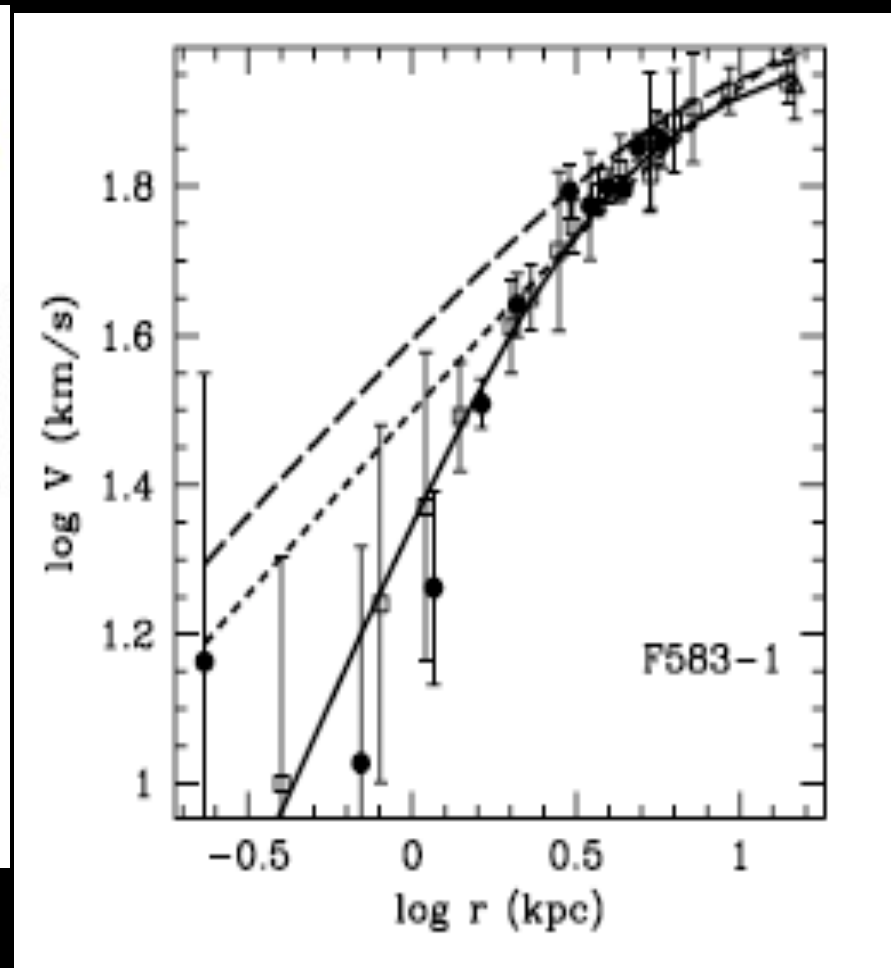
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Low-mass Spirals
($\sim 10^{11} M_\odot$ halos)



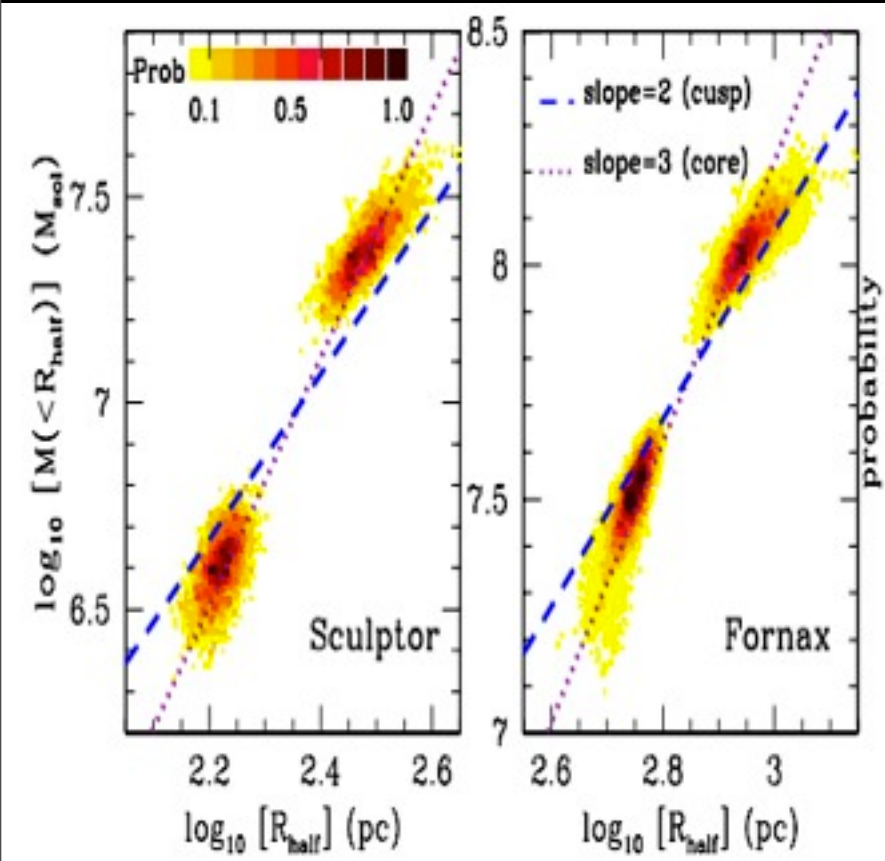
Dwarf core problem

(Kuzio de Naray+ 2008)

Observed cores in ~ 1 – 5 kpc

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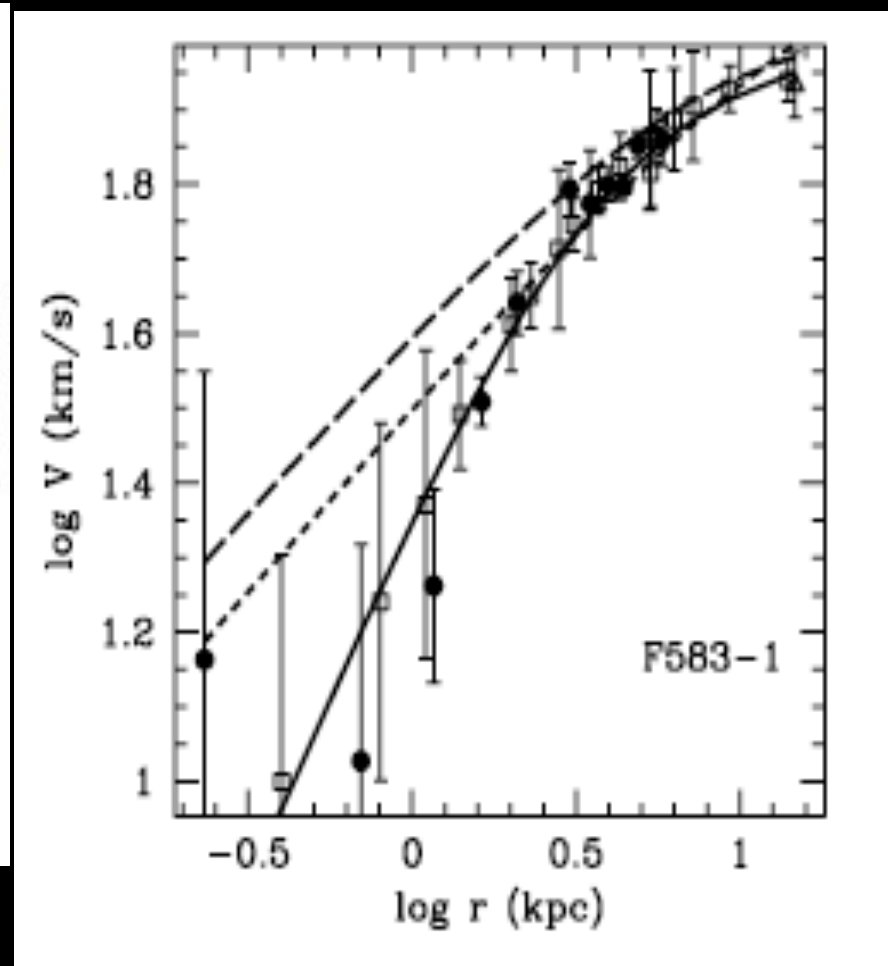
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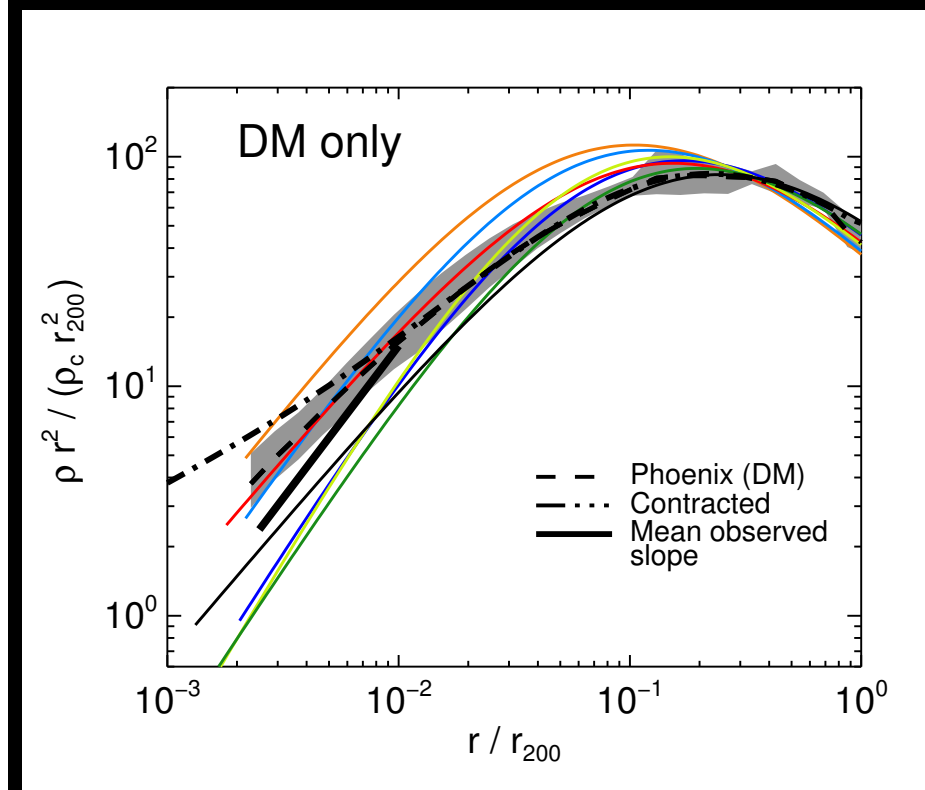


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Galaxy Clusters
($\sim 10^{14}$ – $10^{15} M_\odot$ halos)



Galaxy cluster densities

$\rho \sim r^{-\beta}$

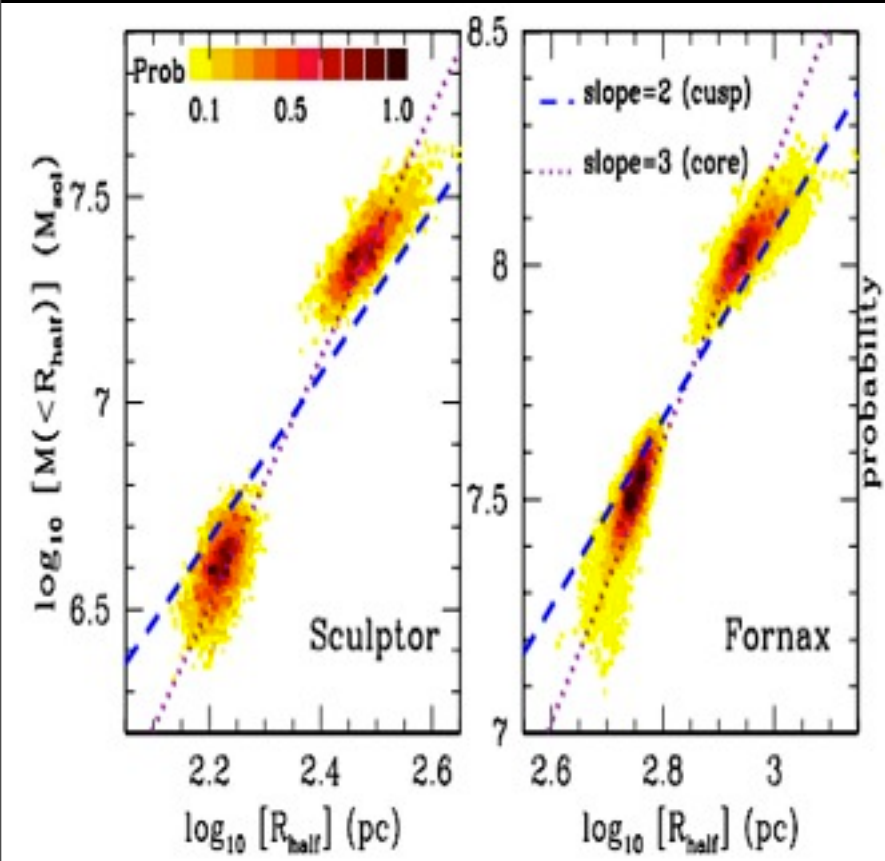
Drew Newman

(Newman+ 2012a,b)

Allow cores of ~ 30 kpc

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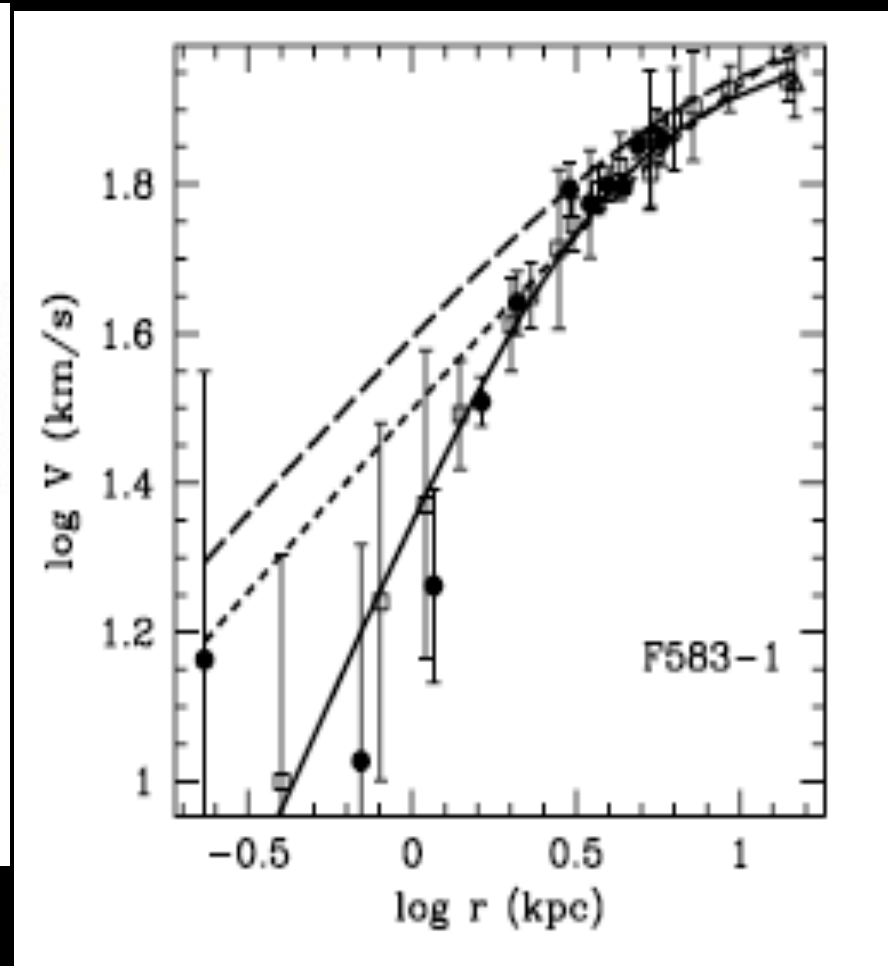
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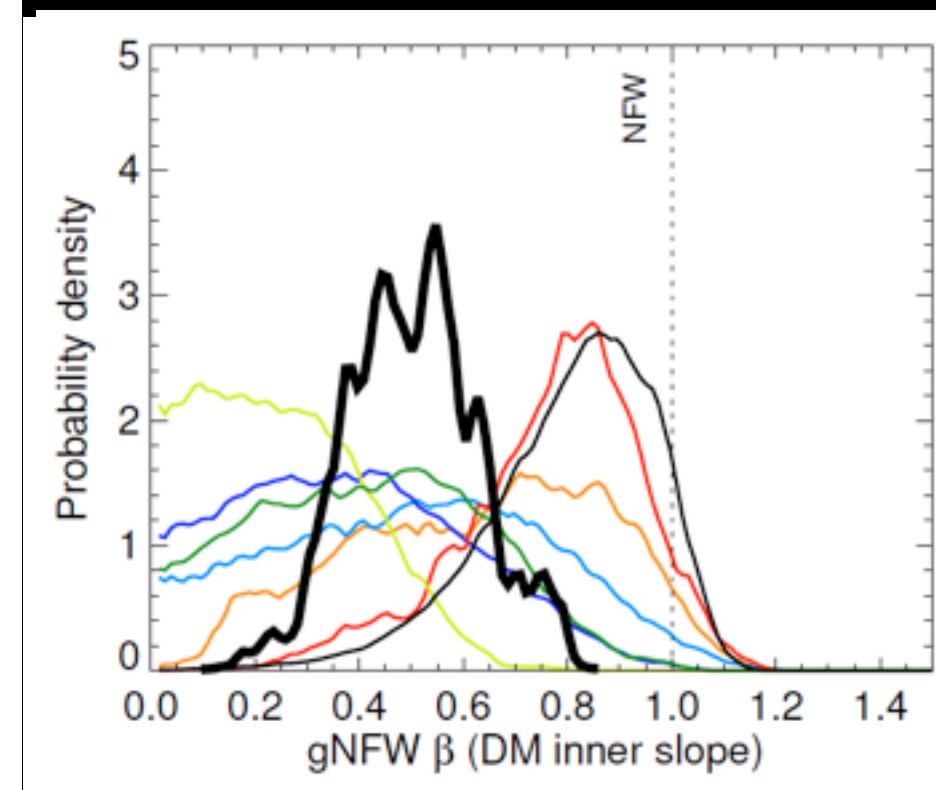


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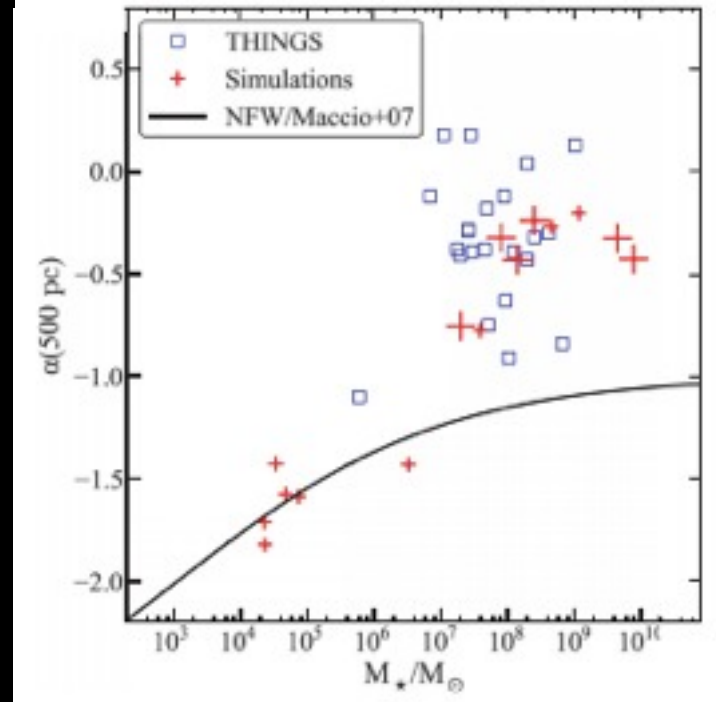
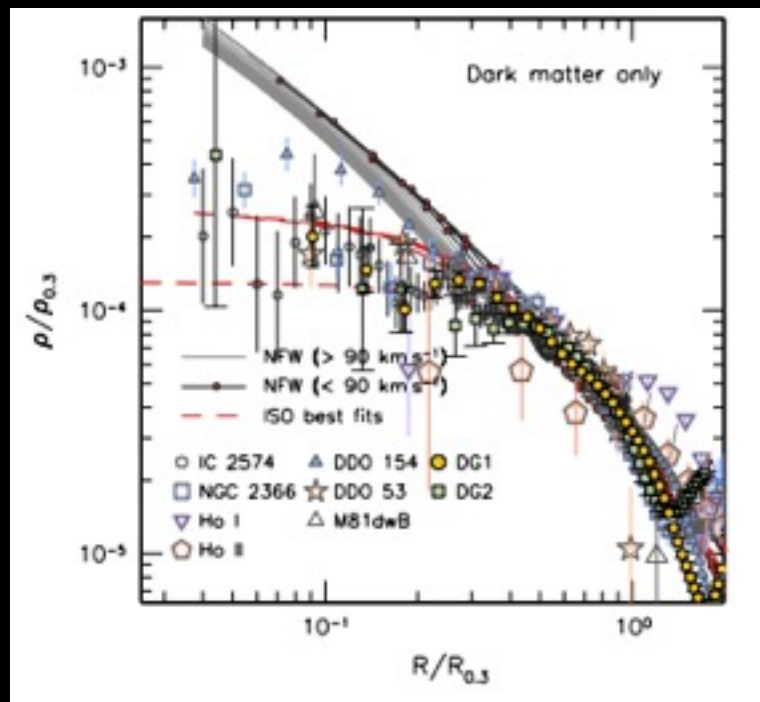
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Are baryons all we need to explain low central densities?

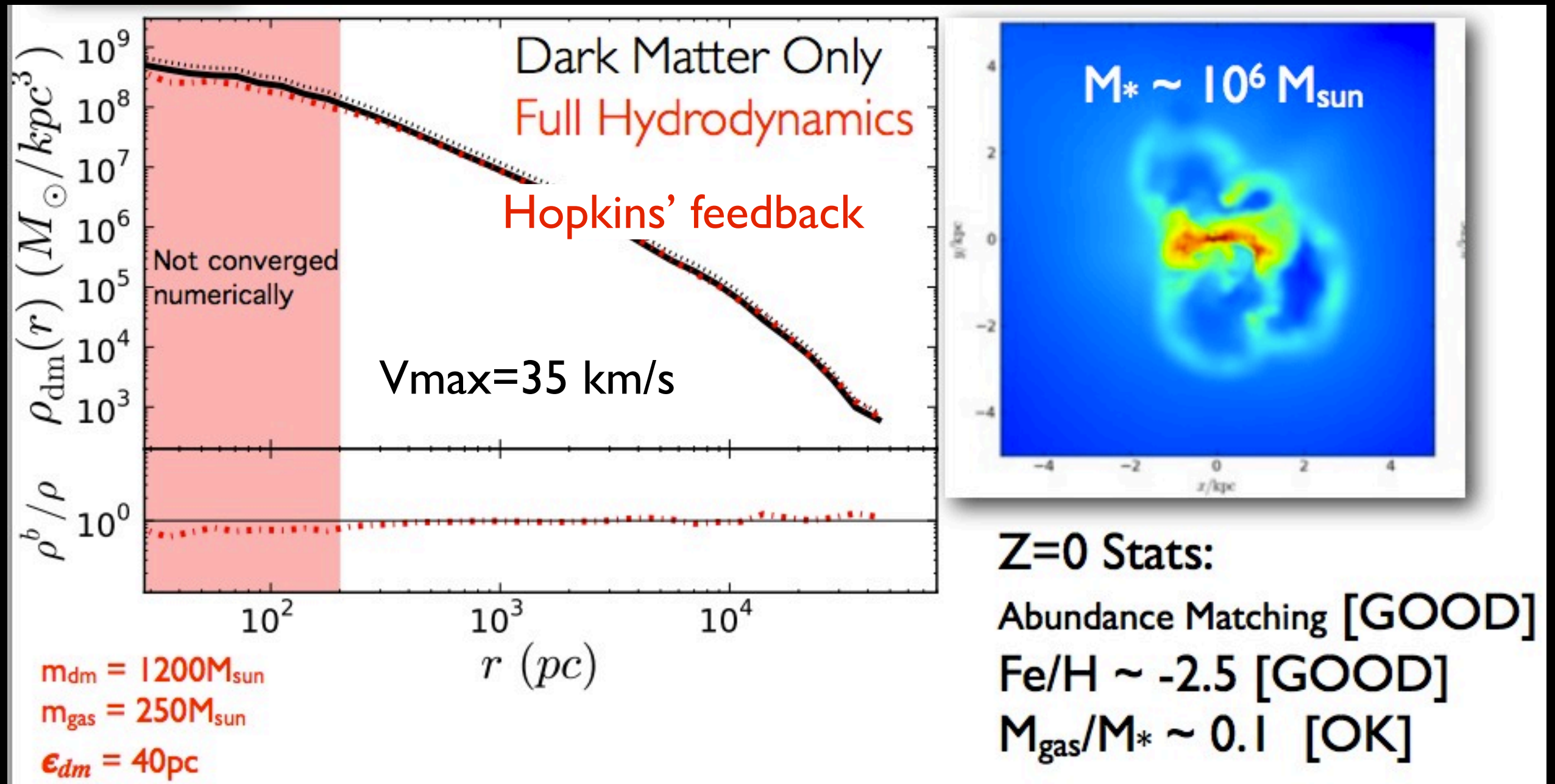


High density thresholds
for star formation
+
turning off cooling a la
Stinson et al. 2006 can
do it

Di Cintio et al. 2013, Zolotov et al. 2012, Brooks et al. 2012, Governato et al. 2012, Teyssier 2012, Pontzen et al. 2012, Oh et al. 2011 (See also Madau in prep.)

- How much is this solution model and resolution dependent?
- What baryonic processes (feedback, tidal interactions, ram pressure, ...) are responsible in transforming the central structure of DM halos at each scale?

Are baryons all we need to explain low central densities?



Oñorbe et al. in prep (See also Trujillo-Gomez in prep)

Other questions we want to answer?

- How much **adiabatic contraction** is there, and is it well described by existing analytical prescriptions? How does this depend on time, and on the mass and accretion history of the halo?
- Is there evidence for a **dark disk**, and how does it depend on merger history?
- Is the **substructure abundance** depleted or enhanced, and how is its distribution in the host halo altered?
- Does baryonic **mass accretion** track DM mass accretion? If not, how and why does it differ?
- How much does baryonic physics alter the **shapes of halos**? As a function of radius?
- Do we see evidence for a **DM density displacement from the galaxies' dynamical center**?

Where possible we will try to directly connect our findings to implications for astronomical observations and DM detection experiments.

Resolution requirements

Observed central densities differ from NFW within:

- $r \sim 500$ pc for $M_{\text{vir}} \sim 10^{10}$ halos
- $r \sim 1\text{-}5$ kpc for $M_{\text{vir}} \sim 10^{11}$ halos
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If we want the density profiles to be resolved to about 30% of these scales, which is:

- $r \sim 100$ pc $\sim 0.05r_s$ for $M_{\text{vir}} \sim 10^{10}$ halos
- $r \sim 0.2\text{-}1$ kpc $\sim 0.05r_s$ for $M_{\text{vir}} \sim 10^{11}$ halos
- $r \sim 1\text{-}2$ kpc $\sim 0.05r_s$ $M_{\text{vir}} \sim 10^{12}$ halos

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✿ **Need $\sim 10^7$ particles within the virial radius to resolve the profiles all the way to $r \sim 0.05 r_s$**

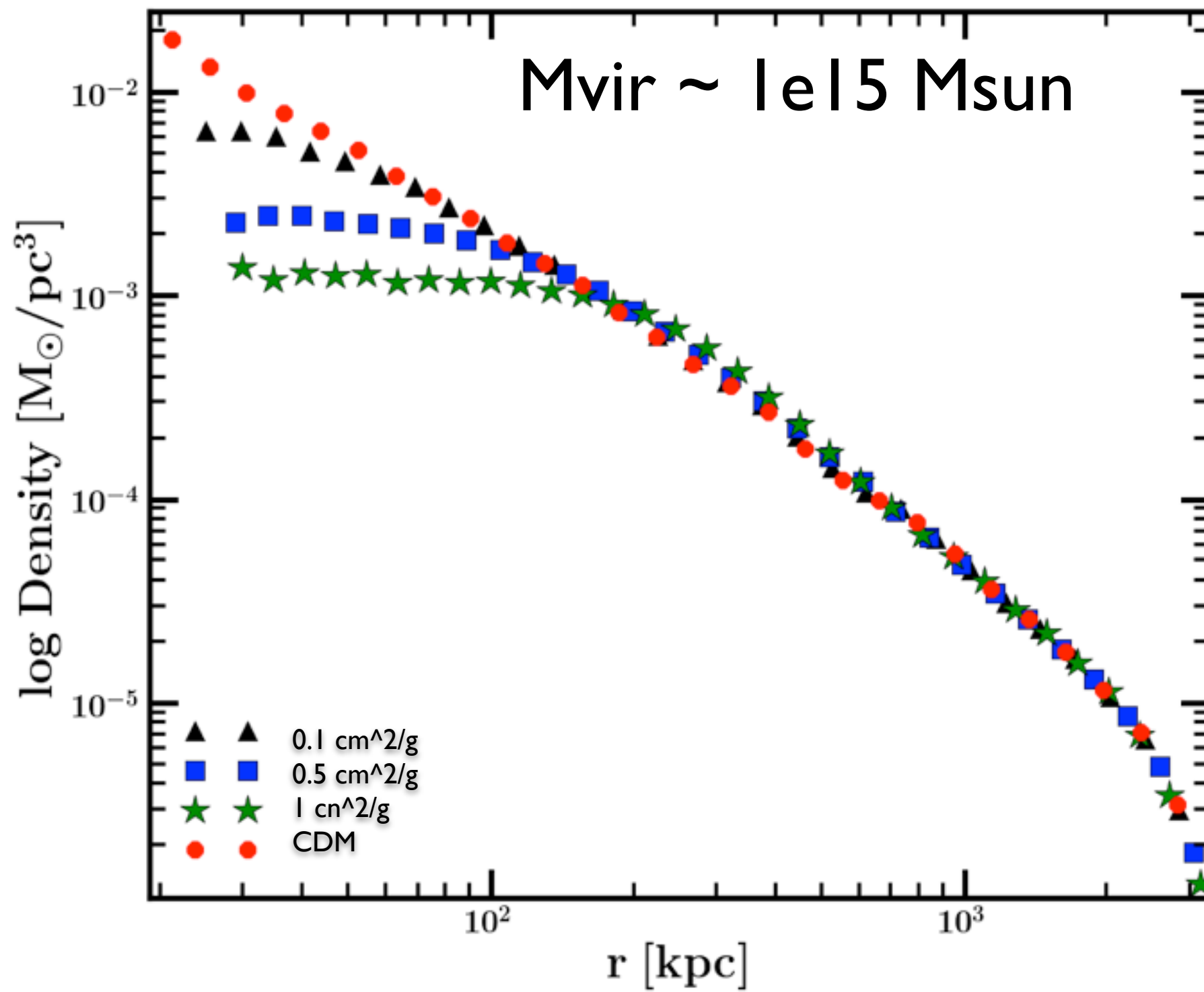
Resolution requirements

Artificial 2-body scattering when $t_{relax} \sim \frac{1}{H_o}$

Power et al. 2003

$$t_{relax} \propto \frac{N_{enc}}{\sqrt{\tilde{\rho}}} \propto \frac{V_{enc}}{m_p} \sqrt{\tilde{\rho}}$$

✿ **Cored halos are harder to resolve!!**



What ICs we need

- $M_{\text{vir}} \sim 1e10$ (quiescent & active), $l_{\text{max}} = 11$ (mp $\sim 1e3$), 12 (mp $\sim 1.3e2$)
 - $M_{\text{vir}} \sim 1e11$ (quiescent & active), $l_{\text{max}} = 12$ (mp $\sim 3.4e5$), 13 (mp $\sim 4e4$)
 - $M_{\text{vir}} \sim 1e12$ (quiescent & active) with $l_{\text{max}} = 12$ (mp $\sim 3.4e5$), 13 (mp $\sim 4e4$), 14 (mp $\sim 5e3$) *
- *Level 14 would be necessary if we want to study subhalos with $M_{\text{vir}} \sim 1e10$ in a MW like host

What Sims we need

List of Planned Simulations

Gadget3 (Resp: Miguel Rocha)

- Dark matter only simulations
- Gadget3 + Hopkins feedback + Hopkins SPH

Gasoline (Resp: ?)

- Dark matter only simulations
- Gasoline + Blastwave

Art-I (Resp: ?)

- Dark matter only simulations
- Art-I + rad. pressure feedback